MIL-R-49122(EL)
1 August 1977

MILITARY SPECIFICATION

RECORDER, SIGNAL DATA RO-481( )/TNS-10

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a recorder for the electrostatic recording of waveforms generated by the sound of artillery fire or shell detonations. An internal power supply provides filtered power and timing pulses. The unit is designated Recorder, Signal Data RO-481( )/TNS-10 (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical (For Use in Electronic Communications and Associated Electrical Equipment)
MIL-S-10956	Sound Ranging Set AN/TNS-10( ), Packaging of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1290

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MIL-P-12749	Paper, Recording, Electrosensitive (Paper PH-438)
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-S-19500	Semiconductor Devices, General Specification for

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-678032	Burn-in Procedure
DL-SM-B-690422	Recorder, Signal Data RO-481( )/TNS-10
SC-A-883793	Performance Test Procedure for Recorder, Signal Data RO-481( )/TNS-10
SC-A-883794	Environmental Test Procedure for Recorder, Signal Data RO-481( )/TNS-10

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 First article. The contractor shall furnish first article units for approval as required by the invitation for bids and contract (see 4.3 and 6.2c).

3.2 Construction. The Recorder, Signal Data RO-481( )/TNS-10 shall be constructed in accordance with the requirements of this specification and of Drawing and Data List DL SM-B-690422 including the requirements for parts, materials and processes thereon.

3.2.1 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.2.2 Paper PH-438. The recorder shall utilize paper PH-438 in accordance with MIL-P-12749. Paper shall be supplied only if and as called for in the solicitation and contract. (See 6.2d)

3.2.3 Cable assemblies. Power and audio-frequency cable and cord assemblies used for voltages up to and including 600 volts AC, or equivalent DC, shall be in accordance with MIL-C-3885.

3.2.4 Semiconductor devices. Semiconductor devices shall be in accordance with Requirement 30 of MIL-STD-454 and with MIL-S-19500. Metallic oxide rectifiers shall not be used.

3.2.5 Systems safety engineering.

3.2.5.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454 (see 4.11).

3.2.5.2 Edge rounding, exposed. All exposed edges and corners shall be rounded (see 4.11).

3.2.5.3 Radioactive materials. Radioactive materials shall not be used (e. g. self luminous dials or markings, electron tubes, surge arrestors and lenses) (see 4.11).

3.3 Performance requirements.

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3.3.1 Power supply.

3.3.1.1 Voltage readings. The microphone voltage (+100 volts) represents the voltage supplied to the Microphones M-161( )/TNS-10. The amplifier voltage (+300 volts) is used to power the recorder amplifiers. The values shown in Table I shall be the minimum output voltages under load, for the listed power supply voltages. The microphone voltage shall not exceed +105 volts. Readings taken at the panel meter shall be within ±5 percent of those taken at the output terminals of the power supply (see 4.6).

Table I. Minimum Voltage Requirements Under Load

Temperature	Battery supply 10.5v 1/		Battery supply 12.5v 1/	
	Amplifier	Microphone	Amplifier	Microphone
23°C ±10°C (73°F ± 18°F)	245	92	290	92
-40°C (-40°F)	245	85	275	85
65.6°C (150°F)	245	95	290	95

1/ Measured at the alligator clips on the battery cable.

3.3.1.2 Power supply ripple. The peak-to-peak ripple voltages, for the microphone supply and the amplifier supply including noise pulses, shall not exceed 2 percent of the values obtained for 3.3.1.1 (see 4.6).

3.3.1.3 Crystal oscillator frequency. The crystal oscillator shall have a frequency of 1000 ±0.1K Hz (see 4.6).

3.3.1.4 Frequency dividers. The frequency of the timing pulse at the 1 Hz output shall be an exact submultiple of the crystal oscillator frequency (see 4.6).

3.3.2 Recorder.

3.3.2.1 Paper speed. The motor shall drive the recording paper at a speed of 15.2 cm (6 inches) per second ±10 percent (see 4.6).

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integrity will be included in the System Engineering Management Plan

3.3.2.2 Pressure plate assembly. The force exerted by the spring on the pressure plate assembly shall be sufficient to keep the recording paper taut, yet shall not restrict it's movement (see 3.3.2.1 and 4.6).

3.3.2.3 Timing marks. The timing marks shall have sharp definition for the full range of input voltages 10.5 - 12.5 volts. There shall be one row of dots every 0.01 second, and an additional row of dots every 0.1 second. Numbers shall be printed sequentially with every tenth 0.1 second dot. The 0.1 second marks shall also be flagged at one second intervals by an additional heavier mark falling in the same row. The corresponding 0.01 and 0.10 second dots shall be so aligned that there will be no error in reading. The distance between any pair of .1 second dots shall not vary by more than 5 percent at any one input voltage (see 4.6).

3.3.2.4 Switching transients. Switching transients caused by operating the test switch shall not cause damage to any portion of the circuitry with a 15.0v supply voltage (see 4.6).

3.3.2.5 Outpost (sensitive) relay. The outpost relay shall close through a line of 1000 ohms resistance, and shall not be closed by the ringing current from Telephone Set TA-312( )/PT when connected to the recorder through Outpost Connecting Box BE-71 (see 4.6).

3.3.2.6 Styli resonant frequency. The stylus assemblies shall have a resonant frequency of 67 ±6 Hz (see 4.6).

3.3.2.7 Communications circuit. The circuit shall provide means for communication between Telephone Set TA-312( )/PT connected to the telephone binding posts and a Telephone Set TA-312( )/PT connected to either a microphone or outpost line when the selector switch is set for that particular line. The ringing signal from the telephone, on any of the lines with 1800 ohms in series, shall cause the associated neon lamps on the operating panel to flash when subjected to the test of 4.6.

3.3.2.8 Sensitivity. The overall sensitivity shall be such as to give a minimum deflection of the styli (peak-to-peak) of 9.5mm (0.375 inch), when a 20 Hz, 0.100 VRMS signal is applied to the microphone terminals (see 4.6).

3.3.2.8.1 Amplifier matching. The sensitivity of the recording channels shall be matched within 6 dB (see 4.6).

3.3.2.9 Distortion. The recorded trace shall be a true representation of the input signal for a total stylus amplitude of 9.5mm (0.375 inch) peak-to-peak (see 4.6).

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3.3.2.10 Starting time. The recorder shall be ready to record in not more than 1.2 seconds after closing the outpost circuit at a battery voltage of 10.5 volts (see 4.6).

3.3.2.11 Accuracy of recording. The recorder shall produce traces such that the variation in time for a signal simultaneously impressed on all channels will not exceed a standard deviation (σ) of 0.002 seconds (see 4.6).

3.3.2.12 Microphone current monitor. The microphone milliammeter shall read 6.1 $\pm$ .5 ma when 10K ohms resitance is placed across the microphone terminals (see 4.6).

3.3.2.13 Attenuator steps. There shall be 16 attenuator steps, with step No 0 as an off position. The attenuation between steps shall be 3 $\pm$0.5 dB for steps 1 thru 15. Between steps 15 and 16 the attenuation shall be 3.5 $\pm$0.5 dB (see 4.6).

3.3.3 Operation. The recording including timing marks shall be clear and legible (see 4.6).

3.4 Finish, protective. The recorder shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel, matching color chip X24087 of FED-STD-595.

3.5 Marking.

3.5.1 General. Marking shall be in accordance with MIL-M-13231 (see 4.4). Front panel marking shall be group I as described in that specification.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the recorder.

3.5.3 Serial numbers. Each recorder shall be serial numbered (see 4.4).

3.6 Burn-in. The recorder shall meet the requirements of this specification after being subjected to the burn-in cycle of Drawing SC-A-678032 (see 4.7).

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3.7 Electromagnetic interference characteristics (EMI). The recorder shall meet the emission requirement RE02 of Table A-II-MIL-I-11748, Class IIIb of MIL-STD-461 (see 4.8).

3.8 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be in accordance with Requirement 7 of MIL-STD-454 (see 4.9).

3.9 Service conditions. After a warmup period of 10 minutes during which the recorder is in "Circuit Test" condition, i.e., all components operational except for the paper drive motor, the recorder shall meet the specified performance requirements of this specification when subjected to the conditions below.

3.9.1 Temperature.

3.9.1.1 Low. The recorder shall meet all the performance requirements of this specification at a temperature of -40°C (-40°F) after being subjected to a non-operating temperature of -45.5°C (-50°F), except that at -40°C (-40°F), the tolerance for paper speed in 3.3.2.1 shall be +10 percent, -30 percent (see 4.10.1.1).

3.9.1.2 High. The recorder shall meet all the performance requirement of this specification at a temperature of 65.6°C (150°F) after being subjected to a non-operating temperature of 71.1°C (160°F) (see 4.10.1.2).

3.9.2 Fungus. The recorder shall be resistant to fungus growth when subjected to fungi under conditions favorable for their development (see 4.10.2).

3.9.3 Altitude (elevation). The recorder shall meet the performance requirements listed below at 3048m (10,000 ft) above sea level after being subjected to a non-operating altitude of 1219m (40,000 ft) above sea level (see 4.10.3).

Paragraph	Measurement
3.3.2.8	Timing marks
3.3.3	Operation

3.9.4 Humidity. The recorder shall be fully operational and free of corrosion when subjected to 94 ± 4 percent relative humidity for extended periods (see 4.10.4).

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3.9.5 Bounce and shock. The recorder shall meet the requirements of Table II (see 4.10.5, 4.10.6 and 4.10.7).

Table II. Bounce and Shock

Requirement	Inspection Para	Performance After Test
Bounce	4.10.5	Full specification performance. Minor mechanical damage is permissible.
Shock; bench handling	4.10.6	No mechanical damage. Hold up hinge shall operate properly.
Shock; drop	4.10.7	Recorder shall meet the operation requirement of 3.3.3. Minor mechanical damage is permissible.

3.10 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
  Slack
  Protection
  Clearance
  Splicing
Connections, general
Grounding, general
Shielding on wire and cable
Soldering and brazing
Cleaning of units
General requirements for securing of parts
Process for welding
Cleaning after welding
Controls
Self-tapping screws
Cleaning prior to welding

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (Does not include packaging.) (See 4.3.)

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.12).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. Unless otherwise specified in the contract, the contractor shall furnish three first article units of the complete Recorder, Signal Data RO-481( )/TNS-10.

4.3.2 First article inspection. This inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, burn-in as specified in 4.7, and the inspections specified for group A, group B, and group C (see Tables IV, V and VI, respectively). First article inspection shall be performed in the order shown in Table III.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

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Table III. First Article Inspection

Inspections	Req Para	Test Para	Unit 1	Unit 2	Unit 3
1. Inspections covered by subsidiary documents 1/	3.4 3.5	4.4	Inspection to be performed on all units		
2. Burn-in 1/	3.6	4.7	Inspection to be performed on all units		
3. Group A inspection 1/	See Table IV		Inspection to be performed on all units		
4. Group B inspection 1/	See Table V		Inspection to be performed on all units		
5. EMI 2/	3.7	4.8		1	
6. Environmental 2/					
Low Temperature	3.9.1.1	4.10.1.1	1		
High Temperature	3.9.1.2	4.10.1.2	2		
Altitude	3.9.3	4.10.3		2	
Humidity	3.9.4	4.10.4	3		
Bounce	3.9.5	4.10.5		3	
Shock; bench handling	3.9.5	4.10.6		4	
Shock; drop	3.9.5	4.10.7		5	
Fungus	3.9.2	4.10.2			1

1/ Inspections 1 to 4 shall be performed, in the order shown, on all first article units before subjecting these units to any other inspection requirements.

2/ The numbers in the unit columns for inspections 5 and 6 specify the order of inspections for the indicated unit.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

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Item	Where required
Parts, materials, and processes	3.2.1
Cable assemblies	3.2.3
Finish	3.4
Marking	3.5
Workmanship	3.10

4.5 Quality conformance inspection of equipment before packaging. Every recorder shall be subjected to burn-in (see 3.6) prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of burn-in shall be classified as a defect and recorded as such.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table IV. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in Table IV.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order satisfactory to the Government, except that the operational inspection of 4.6 shall be last.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots and samples that have been subjected to and met the Group A inspection. This inspection shall conform to Table V and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. The Group B sampling plan, for the AQLs listed in Table V, shall be as follows:

AQL	Inspection level
4%	S-4
6.5%	S-4

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Table IV. Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and mechanical				
Power unit	3.10	4.11		
Amplifier units	3.10	4.11		
Front panel	3.10	4.11	1.0% for	4.0% for
Case, recorder, wood	3.10	4.11	the group	the group
Recorder, complete assy	3.10	4.11		
Stylus assembly	3.10	4.11		
Electrical				
Voltage readings	3.3.1.1	4.6		
Power supply ripple	3.3.1.2	4.6		
Paper speed	3.3.2.1	4.6		
Pressure plate assy	3.3.2.2	4.6		
Timing marks	3.3.2.3	4.6		
Switching transients	3.3.2.4	4.6		1/
Output (sensitive) relay	3.3.2.5	4.6		
Styli resonant frequency	3.3.2.6	4.6		
Communication circuit	3.3.2.7	4.6	1% for	
Sensitivity	3.3.2.8	4.6	the group	
Amplifier matching	3.3.2.8.1	4.6		
Distortion	3.3.2.9	4.6		
Starting time	3.3.2.10	4.6		
Accuracy of recording	3.3.2.11	4.6		
Microphone current monitor	3.3.2.12	4.6		
Operation	3.3.3	4.6		

1/ Defects in all characteristics other than visual and mechanical are considered major.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order satisfactory to the Government.

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Table V. Group B Inspection

Inspection	Req Para	Insp Para	AQL
Crystal oscillator frequency	3.3.1.3	4.6	4% for the group
Frequency dividers	3.3.1.4	4.6	
Attenuator steps	3.3.2.13	4.6	
Interchangeability	3.8	4.9	6.5%

4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units of product that have been subjected to and passed Group A and Group B inspection requirements. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. Three sample recorders shall be selected at random, one sample for each Subgroup in Table VI, from the first twenty units or fraction thereof produced. The EMI and fungus tests shall be performed only on the first production lot. Thereafter, two units of the recorder shall be selected at random, one sample for subgroups 1 and 2 of Table VI from every twenty units or fraction of at least ten units produced. If the fraction is less than 10, the selection of Group C samples for the preceding test shall include units from the remaining fraction.

4.5.3.1.1 Sampling for inspection of solderless-terminal-lug connections. Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1-foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government except that the fungus test (4.10.2) shall be last.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group B and Group A inspection in that order.

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Table VI. Group C Inspection

Inspection	Req Para	Insp Para
Subgroup 1		
Low temperature	3.9.1.1	4.10.1.1
High temperature	3.9.1.2	4.10.1.2
Humidity	3.9.4	4.10.4
Subgroup 2		
Electromagnetic interference characteristics 1/	3.7	4.8
Altitude	3.9.3	4.10.3
Bounce	3.9.5	4.10.5
Shock; bench handling	3.9.5	4.10.6
Shock; drop	3.9.5	4.10.7
Subgroup 3		
Fungus 1/ 2/	3.9.2	4.10.2

1/ Electromagnetic interference characteristics and fungus tests shall only be performed during first article testing and on the first production lot.

2/ The recorder shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.6 Performance tests. The performance tests of Recorder, Signal Data RO-481( )/TNS-10 shall be performed as specified in the test procedure, Drawing SC-A-883793.

Requirement	Paragraph
Voltage readings	3.3.1.1
Power supply ripple	3.3.1.2
Crystal oscillator frequency	3.3.1.3
Frequency dividers	3.3.1.4
Paper speed	3.3.2.1
Pressure plate assembly	3.3.2.2

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Requirement	Paragraph
Timing marks	3.3.2.3
Switching transients	3.3.2.4
Outpost (sensitive) relay	3.3.2.5
Styli resonant frequency	3.3.2.6
Communications circuit	3.3.2.7
Sensitivity	3.3.2.8
Amplifier matching	3.3.2.8.1
Distortion	3.3.2.9
Starting time	3.3.2.10
Accuracy of recording	3.3.2.11
Microphone current monitor	3.3.2.12
Attenuator steps	3.3.2.13
Operation	3.3.3

4.7 Burn-in. All units submitted for inspection shall be subjected to burn-in in accordance with Drawing SC-A-678032 (see 3.6).

4.8 Electromagnetic interference compatibility. The recorder shall be tested for conformance to test method RE02 of Table AII-MIL-I-11748, Class IIIb of MIL-STD-461 (see 3.7).

4.9 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

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4.10 Service conditions. The Recorder, Signal Data RO-481( )/TNS-10 shall be tested in accordance with the following conditions:

4.10.1 Temperature test.

4.10.1.1 Low. The RO-481( )/TNS-10 shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810 as specified on Drawing SC-A-883794 (see 3.9.1.1).

4.10.1.2 High. The RO-481( )/TNS-10 shall be subjected to the test of Method 501.1, Procedure II, of MIL-STD-810 as specified on Drawing SC-A-883794 (see 3.9.1.2).

4.10.2 Fungus test. The RO-481( )/TNS-10 shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810 as specified on Drawing SC-A-883794 (see 3.9.2).

4.10.3 Altitude (elevation) test. The RO-481( )/TNS-10 shall be subjected to the test of Method 500.1, Procedure I, of MIL-STD-810 as specified on Drawing SC-A-883794 (see 3.9.3).

4.10.4 Humidity test. The RO-481( )/TNS-10 shall be subjected to the test of Method 507.1, Procedure II, of MIL-STD-810 as specified on Drawing SC-A-883794 (see 3.9.4).

4.10.5 Bounce test. The RO-481( )/TNS-10 shall be subjected to the test of Method 514.2, Procedure XI, of MIL-STD-810 as specified on Drawing SC-A-883794 (see 3.9.5).

4.10.6 Shock test; bench handling. The RO-481( )/TNS-10 shall be subjected to the test of Method 516.2, Procedure V, of MIL-STD-810 as specified on Drawing SC-A-883794 (see 3.9.5).

4.10.7 Shock test; drop. The RO-481( )/TNS-10 shall be subjected to the test of Method 516.2, Procedure II, of MIL-STD-810 as specified on Drawing SC-A-883794 (see 3.9.5).

4.11 Visual and mechanical inspection. The recorder shall be examined for the defects listed in MIL-STD-252 and Table VII, and inspected to verify conformance to those portions of 3.2.5 which can be determined visually.

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terminals P and * digital signal and generates the output for

Table VII. Classification of Visual and Mechanical Defects

Classification	Defect
Major	Improper or lack of tropicalization. Carrying case too large or too small; compartments wrong size. Improper or loose fitting panels and covers Controls loose or too tight. Pen action not smooth due to mechanical defects. Improper meshing and/or loose gears. Paper does not unroll smoothly and evenly. Improper or uneven pressure applied by pressure rollers. Components not interchangeable. Loose binding posts, cable plugs, and/or jacks. Stylus of stylus assembly not properly secured or not at correct angle
Minor	Loose fasteners. Shock mounts improperly mounted, or loose. Viewing window misaligned and/or distorted. Markings missing, incomplete and/or illegible. Loose rollers. Loose cabling. Rough edges on brazed, soldered and/or welded joints. Flux residue. Improper finish.

4.12 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-S-10956 to determine conformance to the requirements of Section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-10956.

6. NOTES

6.1 Intended use. Recorder, Signal Data RO-481( )/TNS-10 is a unit of Sound Ranging Set AN/TNS-10( ) which is used to detect and locate the source of the sound of artillery fire or shell detonations.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level of packaging and level of packing required for shipment (Level A, Level B, or Commercial).

c. Number of first articles to be submitted for approval (see 4.3).

d. Type and quantity of paper required (see 3.2.2).

e. Quantity of tools required.

f. The following equipments should be loaned to the successful bidder for testing as indicated in the bid request and contract:

2 ea Telephone Set TA-312( )/PT
1 ea Outpost Connecting Box BE-71
1 ea Outpost Switch Unit SA-45/G

g. Submission of statement of treatment before first article approval. This statement should be submitted to the contracting officer.

6.3 Nomenclature. The parentheses in the nomenclature of the equipment will be deleted or replaced by a letter indicating the particular design; for example Recorder RO-481K/TNS-10. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Custodian:
Army-EL

Preparing Activity:
Army-EL

Project No 1290-A335

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-R-49122(EL) RECORDER, SIGNAL DATA RO-L81( )/TNS-10

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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SUBMITTED BY *(Printed or typed name and address — Optional)*

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DATE

DD FORM 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

lined in paragraph 1.2.1.3.4.

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